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# ***Headquarters U.S. Air Force***

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## **Runway Incursion Prevention**



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HQ AFFSA/A3AV  
17 March 09  
Version 1

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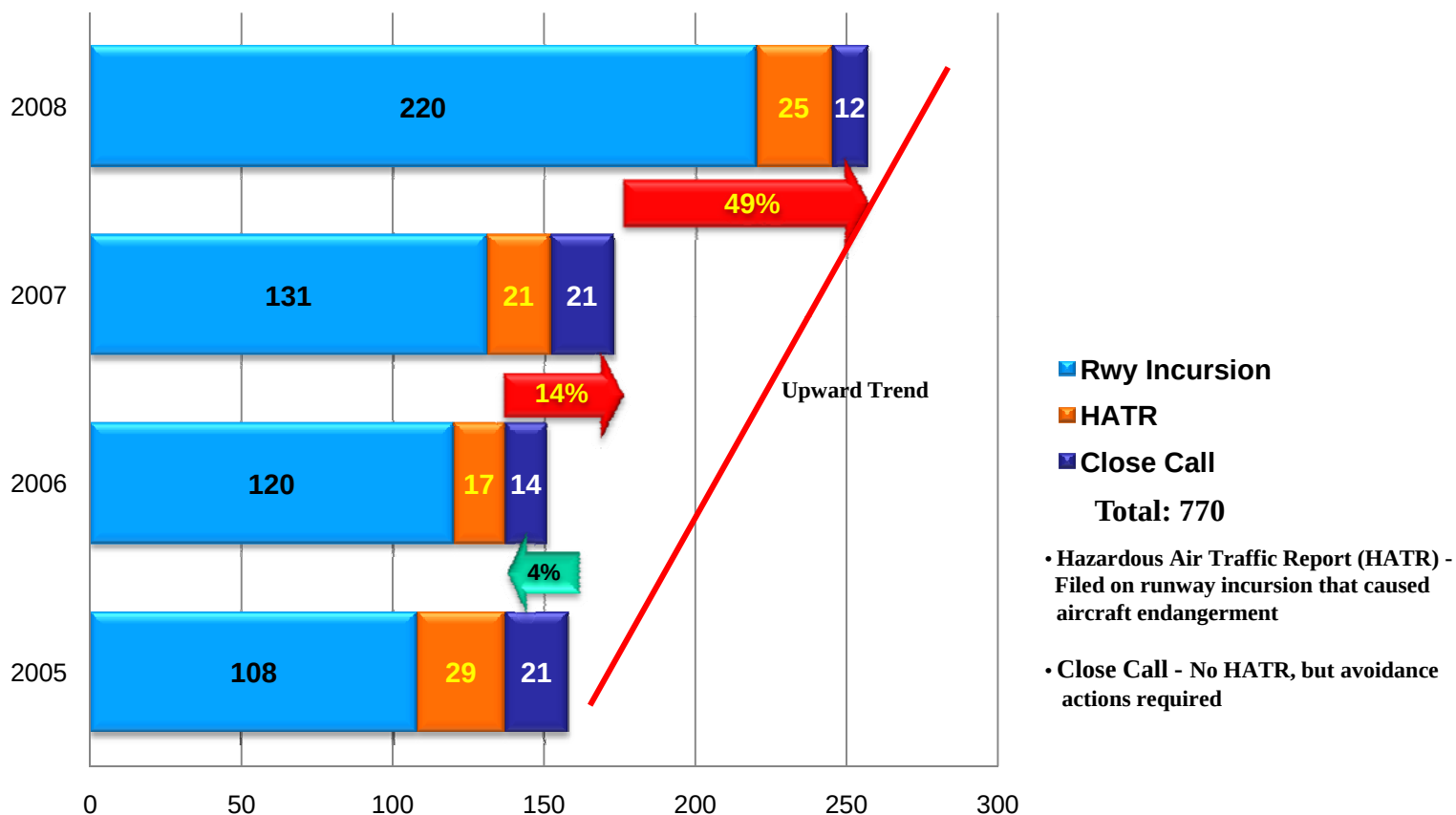
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# Runway Incursion Trends

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Data Source: AF Safety Automated System (AFSAS) (As of 13 Jan 09)

**CY08 runway incursions exceeded CY07 by 49%**  
**CY 05-08, 16% of runway incursions filed as HATRs**



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# Overview

- Definitions
- Runway Incursion Trends
- Recent Examples
- Common Causes
  - Driver Training
  - Situational Awareness
  - Communication
- Ways to reduce runway incursions
- What you should know now
- Local Airfield Information





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# Definitions

- **Controlled Movement Area (CMA):** “Any portion of the airfield requiring aircraft, vehicles and pedestrians to obtain specific ATC approval for access.”
- **Controlled Movement Area Violation (CMAV):** “An airfield infraction caused by aircraft, vehicles, or pedestrians entering the controlled movement area without specific ATC approval.” This also includes runway incursions, the most serious form of CMAVs.



**Reference: AFI 13-213, Airfield Management**

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## ***Definitions***

- **Recommend adding diagram that depicts CMAs at local airfield**



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# *Runway Incursion Trends*

- Runway incursions Categories (Vehicle/Pilots/Pedestrian/ATC)
  - 87% Vehicle/Pedestrian
    - 91% were Trained Individuals
    - 9% by Pilots
    - 4% by ATC (Operational Errors)
- Aggressive actions required to lower trend and decrease risks on airfields:
  - Improve vehicle driver training
  - Increase situational awareness and tracking of airfield vehicles



**91% of airfield drivers who committed a runway incursion were trained**

Data provided by AF Safety Automated System (AFSAS) and is current as of 13 Jan 09



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# Runway Incursion Example #1



An untrained vehicle passenger requested permission from ATC to cross the wrong runway. ATC cleared two F-15s for takeoff and they overflew the vehicle.

Data Source: AFSAS

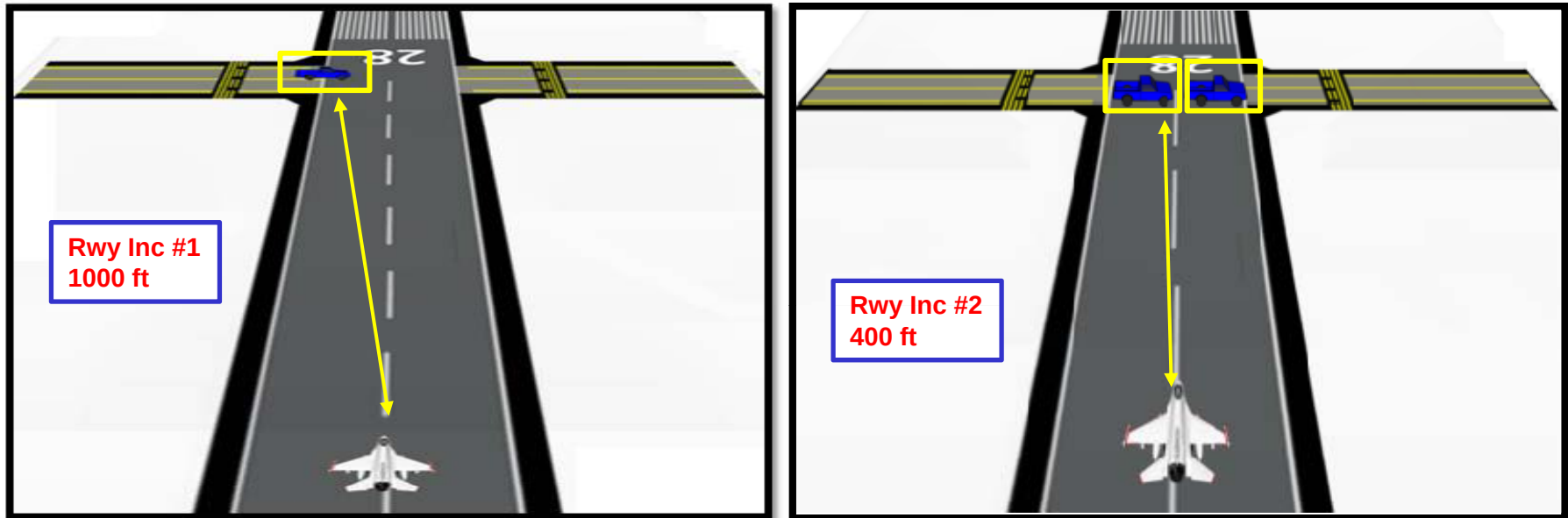
**Know where you are at all times**



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## Runway Incursion Example #2

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During an exercise, a vehicle crossed the runway without approval from ATC and passed within 1000 ft of an F-16 on landing rollout. Seconds later two more vehicles crossed the same runway without ATC approval and passed within 400 ft of the same F16 on landing rollout.

Data Source: AFSAS

**WARNING: An exercise, aircraft emergency, or incident response does not authorize a vehicle to enter the runway without approval from ATC**





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# ***Common Causes***

- **Poor Training**
  - **Inadequate training program**
  - **Not fully comprehending airfield driving training concepts**
  - **Not correctly applying operating procedures**
  - **Drivers not accomplishing annual refresher training**



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# ***Driver Training***

## **Airfield Driving Program Responsibilities**

### **Wing/Group Leadership**

- **Designate personnel and agencies to support program**
- **Approve publication of Airfield Driving Instruction**

### **Squadron Commanders**

- **Appoint Airfield Driving Program Manager/Alternate**
- **Ensure assigned personnel complete training**
- **Limit number of airfield drivers to absolute minimum**

**Squadron commanders are ultimately responsible to ensure their people are adequately trained and follow published procedures**



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# ***Driver Training***

## **Airfield Driver Training Responsibilities**

### **Airfield Management**

- Wing Manager for Airfield Driving
- Trains all ADPMs
- Conducts Staff Assistance Visits
- Provide quality control enforcement
- Certify airfield drivers

### **Airfield Driving Program Manager**

- Train all unit airfield driver trainers
- Validate drivers training complete
- Ensure annual refresher training conducted/annotated on AF IMT 483

### **Airfield Driving Trainer**

- Conduct required training
- Ensure driver is qualified to drive on airfield

### **Airfield Driver**

- Complete all required training
- AF Airfield Driving CBT
- Airfield Layout test
- Color vision test
- Airfield tour
- Practical test of driving on the airfield
- Complete phraseology training
- Local airfield driver's test
- Maintain proficiency

**Effective airfield driver's training is everyone's responsibility**

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# ***Driver Training***

- **Did you successfully complete...**
  - **Air Force Airfield Driving Computer Based Training (CBT)?**
  - **Local airfield driving procedures training?**
  - **Practical day and night driving test?**
  - **Local airfield driver's test?**
  - **Airfield layout and Light Gun Signal Recognition Test?**
  - **Color vision test?**
  - **Phraseology test (if operating in CMA)?**
  - **Annual refresher training?**

**Completion of all training requirements are mandatory**



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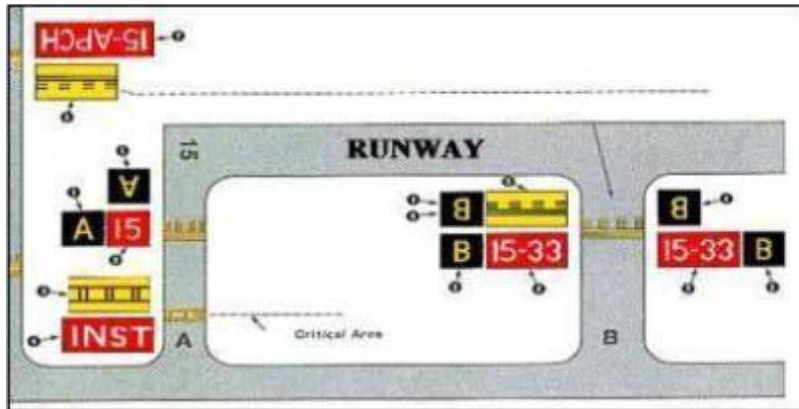
# ***Common Causes***

- **Poor Training**
  - Inadequate training program
  - Not fully comprehending airfield driving training concepts
  - Not correctly applying operating procedures
  - Drivers not accomplishing annual refresher training
- **Poor Situational Awareness**
  - Unfamiliar with airfield or complex configurations
  - Inexperienced drivers driving on the airfield
  - Drivers assuming authorization to enter runway in response to aircraft emergencies



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# Situational Awareness



**Airfield Layout, Signs, Lights, and Markings:** Know your airfield layout. Know what airfield signs, lights and markings mean and what action to take.

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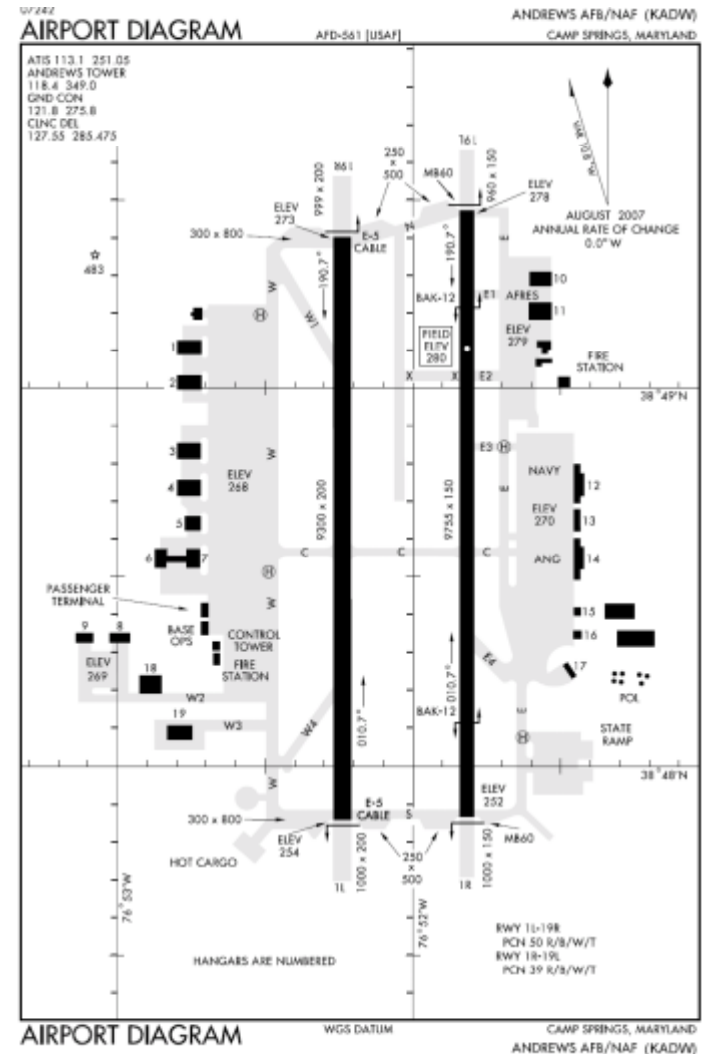


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# Situational Awareness

- **Know Your Location:** Use a current airfield diagram and advisories for airfield activities
- **Limit Distractions:** Focus on surroundings and ongoing activities especially when operating near runways and aircraft
- **Follow Guidance:** Adhere to ATC instructions and airfield signs and markings
- **Admit When Lost:** Find a safe place to stop, then ask Tower for help

**Keep a current airfield diagram in the vehicle**





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# Situational Awareness



AF Visual Aid 11-240, Airport Signs and Markings



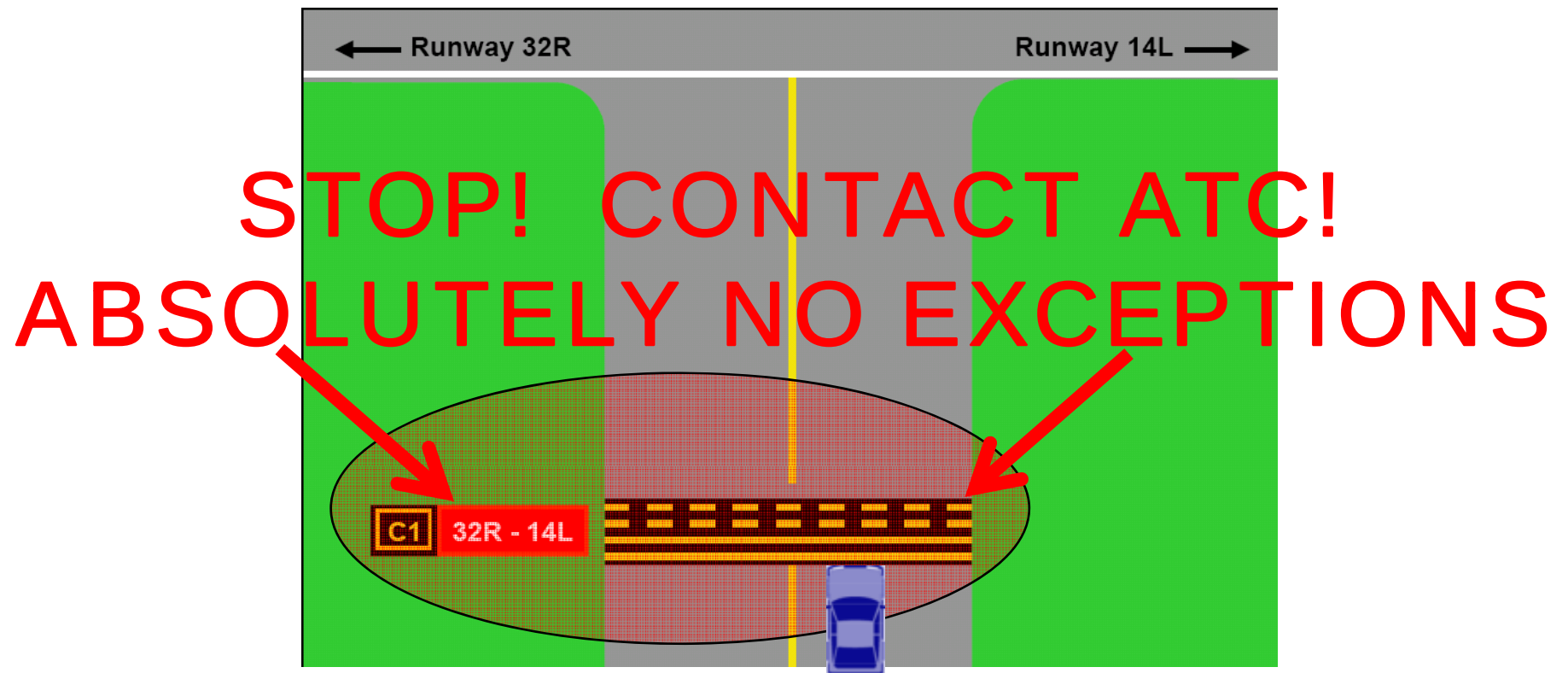
**Ensure the Airport Signs and Markings Placard is in all vehicles that operate on the airfield.**





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# *Situational Awareness*



**Always stop and contact ATC at the runway hold sign and marking**

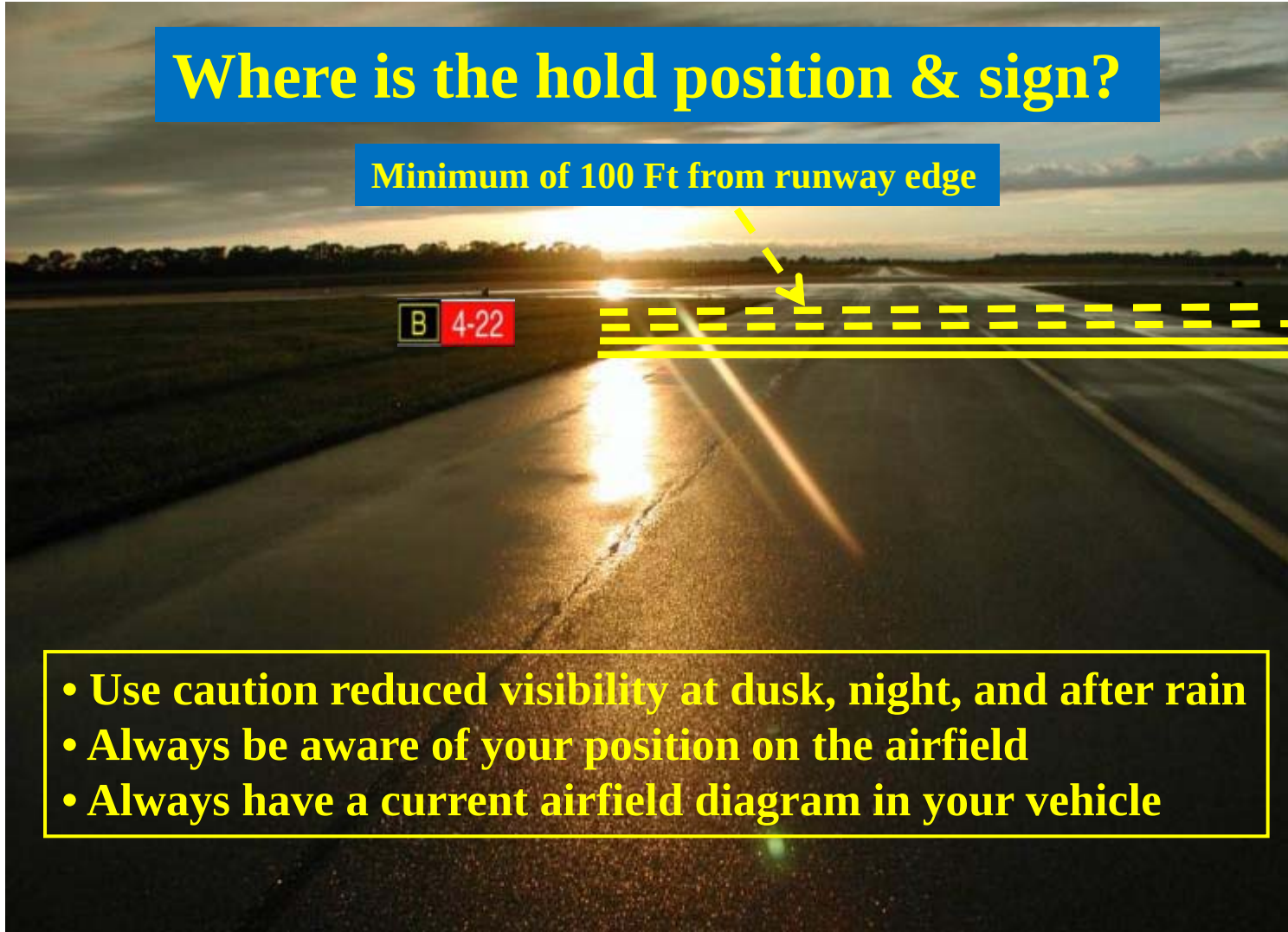


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# *Situational Awareness*

**Where is the hold position & sign?**

**Minimum of 100 Ft from runway edge**



- Use caution reduced visibility at dusk, night, and after rain
- Always be aware of your position on the airfield
- Always have a current airfield diagram in your vehicle

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# ***Situational Awareness***

- **Activate vehicle headlights, flashers, and/or light bar when in a controlled movement area or on the runway during periods of reduced visibility caused by weather or light conditions**
- **Helps ATC and others identify your vehicle location and movement**



**Make sure ATC and others see you at all times**

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# ***Situational Awareness***

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- Emergency and incident responses do not change the rules. ATC permission is required to enter the runway
- Do not follow other emergency response vehicles on the runway unless you have pre-arranged procedures for escort/convoy
- ATC approval is required to enter the runway at all times, even during suspended/closed ops
- Place entry control points outside runway hold lines unless coordinated with the on-scene commander and ATC



**Aircraft emergencies, exercises, or incident responses do not authorize you on runway**



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# ***Common Causes***

- **Poor Training**
    - Inadequate training program
    - Not fully comprehending airfield driving training concepts
    - Not correctly applying operating procedures
    - Drivers not accomplishing annual refresher training
  - **Poor Situational Awareness**
    - Unfamiliar with airfield or complex configurations
    - Inexperienced drivers driving on the airfield
    - Drivers assuming authorization to enter runway in response to aircraft emergencies
  - **Poor Communication**
    - Misunderstanding ATC instructions
    - Vehicle call sign mix up
    - Anticipating ATC approval to cross/enter runway
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# ***Communications***

- **Never assume:**
  - **ATC knows who you are, where you are, or what you are doing**
  - **You have approval to enter the runway unless you receive a clear radio transmission from ATC**
  - **You know more than ATC about what aircraft are landing, taking off, or flying in the local area**
  - **The runway is closed unless verified through ATC or Airfield Management**



**Always get ATC approval prior to entering the runway environment!**





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# Communications

- **Maintain the “Big Picture”:** Monitor the radio when operating on runway and taxiways
- **Transmit Clearly:** Plan your transmission and pay attention to standard phraseology
- **Listen Carefully:** Listen to ATC and other transmissions to increase situational awareness, don't let communications become automatic
- **Make Sure You are Heard:** Test your radio before entering the airfield and be aware of transmission blind spots on the airfield
- **Proper Equipment:** Due to handheld radio's limited range and battery failures install mobile radios in vehicles that frequently access runways



**Effective communication is the key to runway incursion prevention**

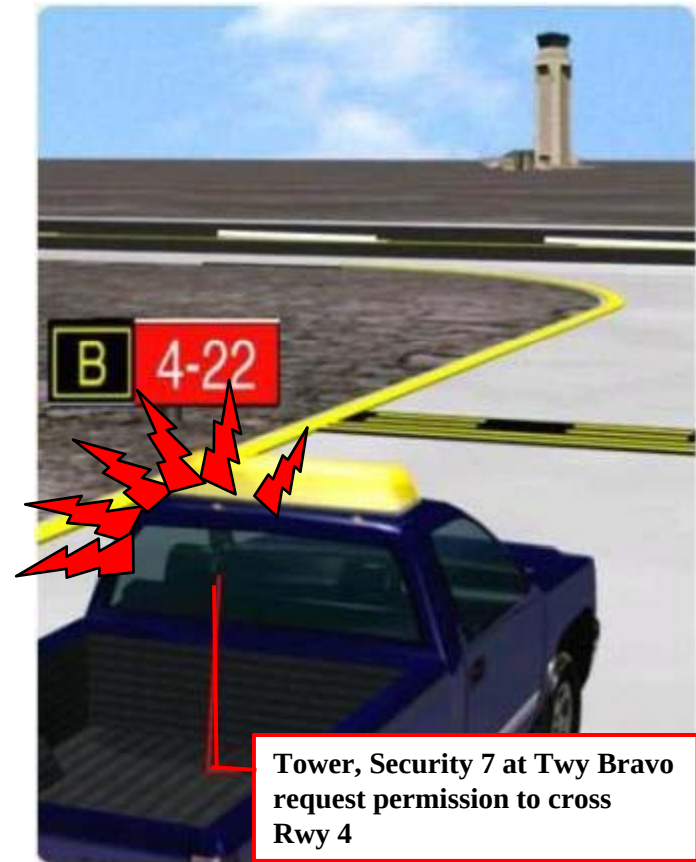
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# Communications

- **Radio Terminology/Phraseology:**
  - Identify yourself with your call sign
  - State location & intentions to ATC
  - Wait for ATC response
  - If you do not understand ATC response, ask them to repeat by saying "SAY AGAIN"



**Never move your vehicle unless you are positive ATC knows what you want and has given you approval to proceed**





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# Communications

## ■ Radio Terminology/Phraseology:

- The only ATC phraseology authorizing you to enter or cross any controlled area is “PROCEED” or “CROSS”
- Acknowledge receipt of ATC instructions by repeating instructions verbatim
- Notify ATC after exiting a CMA until they acknowledge the transmission

Security 7, Tower,  
Proceed across Rwy 4,  
Report when off



**Always follow ATC instructions and readback verbatim**



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# ***Ways to Reduce Incursions***

- **Ensure the Airfield Driving program meets AFIs/guidance**
- **Enforce compliance with local airfield driver instructions**
- **Minimize the number of drivers authorized on the airfield**
- **Increase Airfield Driver's Certificate spot checks**
- **Identify and fix radio communication problems with vehicles**
- **Minimize runway crossings during increased aircraft operations**
- **Improve training on runway crossing procedures/phraseology**
- **Improve airfield signage and markings**
- **Develop local area information pamphlets on "high threat" areas**
- **More training emphasis on controlled movement area access procedures**

**We need your help to reduce runway incursions**



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# ***What You Should Know Now***

- **Know where you are at all times**
- **Never move your vehicle unless ATC has given you approval to enter the controlled movement area**
- **Make sure ATC and others see you at all times**
- **Always follow ATC instructions and readback verbatim**
- **Aircraft emergencies, exercises, or incident responses require ATC approval to enter the runway**

**Stay alert, follow procedures, and always drive safely**



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# ***Local Base Information Slides***

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- Possible topic areas to consider include:
  - Airfield driver training and certification procedures
  - Key airfield driving points of contact
  - Location of controlled movement areas
  - Unique airfield operating procedures
  - Confusing or complex areas or signage
  - ATC visual blind spots
  - Radio transmission blind spots
  - Recognized “high threat” areas on the airfield
  - Recommend using digital pictures of the actual areas on the airfield to help present the information